Phase 2: Ratio Decidendi Extraction and Plain-Language Rewriting

Project: Multilingual Courtroom Judgment Simplifier

Objective: To extract the ratio decidendi (the legal principle on which the court decided the case) from a judgment automatically, measure how close it is to the reference ratio in the dataset, and rewrite it in plain language.

1. Setup

- Model: Qwen2.5-3B-Instruct on a Google Colab T4 GPU, the same model as the Phase 1 baseline. Generation was deterministic (no sampling), so results can be repeated.
- Input: the facts, issues, arguments and final order of each case. The Ratio_Decidendi column was deliberately kept out of the input and used only as the reference answer for scoring.
- Test data: the first 20 cases of the 200-case test set. Both versions below were run on the same 20 cases, so the comparison is fair.
- All predictions were saved to Google Drive after each case (outputs folder), so work can resume after a disconnection.

2. Method

Two prompting approaches were compared.

Version 1: plain prompt. The model was told it is an expert in Indian civil law and asked to state the ratio decidendi in 3 to 4 sentences, mention the relevant legal rule or section and why it decided the outcome, and not to retell the facts (maximum 300 new tokens).

Version 2: prompt with two worked examples (few-shot). Two complete cases from the training set, each with its real ratio, were shown to the model before the test case. The instruction also asked it to start directly with the legal principle (maximum 350 new tokens). The aim was for the model to copy the style, length and statute citations of a real ratio.

3. Evaluation

- ROUGE-1 and ROUGE-L: overlap of words and phrases with the reference ratio. These are strict, because a correct idea in different words scores low.
- BERTScore F1: similarity of meaning, which is a fairer measure for this task.
- Manual comparison of a predicted ratio with the reference, to check which legal principles were captured and whether anything was invented.

4. Results

Table 1: Average scores on 20 test cases

Version	ROUGE-1	ROUGE-L	BERTScore F1
v1: plain prompt	0.402	0.208	0.844
v2: two examples	0.397	0.229	0.858

Version 2 improved BERTScore by about 0.014 and ROUGE-L by about 0.02, while ROUGE-1 was almost unchanged (a difference of 0.005, which is within normal variation). With only 20 cases, this is a modest and suggestive improvement, not a proven one.

ROUGE scores are low for both versions because the reference ratios are long and formal, and the model expresses the same ideas in different words.

5. Case study: Meghdoot Agro Industries vs. Padmavathi Gowda and Others

> **Reference ratio:** An easement of necessity under Section 13 of the Indian Easements Act, 1882 arises only upon severance of a tenement and is confined to what is absolutely necessary for enjoyment of the dominant heritage; mere convenience is not necessity. A prescriptive easement under Section 15 requires user as of right (nec vi, nec clam, nec precario), and any admission of permissive user destroys the very foundation of the claim. Existence of an alternative, even if longer and less convenient, negatives necessity. The rule that he who seeks equity must do equity applies with full force to a party invoking the discretionary jurisdiction of the Court. Interest is compensatory and not penal, and the rate awarded must bear a reasonable relationship to prevailing commercial rates.
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> **Version 1:** The ratio decidendi of this judgment is that the court found a limited easement of necessity rather than an easement by prescription. The court held that while the plaintiff had used the track openly and peaceably for over twenty years, the user was not "as of right" but with the permission of the defendant's father, which is inconsistent with an easement by prescription. Therefore, the court did not find an easement by prescription, but instead declared a limited easement of necessity for the purposes of ingress and egress, allowing the plaintiff to use a strip of eight feet in width. This decision was based on the principle that where a person's land is landlocked and there is no reasonable alternative access, an easement of necessity can be created to allow passage through another's land. The court also considered the plaintiff's lack of an alternative access route, which supported the creation of such an easement.
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> **Version 2:** An easement by prescription under Section 15 of the Indian Easements Act, 1882, requires continuous and uninterrupted use as of right for twenty years ending within two years prior to the suit. In cases of necessity, however, a shorter period may suffice, and the court can grant an easement if the user is necessary for the enjoyment of the dominant tenement. Here, the plaintiff proved an easement of necessity due to the land's inherent landlocked condition, necessitating an alternative route for ingress and egress.

Table 2: Which principles of the reference ratio were captured

Principle in the reference ratio	Reference	Version 1	Version 2
Section 13 (easement of necessity) cited	Yes	No	No
Section 15 (prescriptive easement) cited	Yes	No	Yes
Necessity separated from prescription	Yes	Yes	Partly
Permissive use defeats a prescriptive claim	Yes	Yes	No
No alternative access supports necessity	Yes	Yes	Partly (landlocked)
Tests: severance, "mere convenience is not necessity", nec vi / nec clam / nec precario	Yes	No	No
Equity and interest principles	Yes	No	No

Version 1 stayed closer to the facts of the case and captured the central distinction, but cited no statute. Version 2 sounds more like a real ratio and cites Section 15, but it lost the point about permissive use and added the statement that "a shorter period may suffice" for an easement of necessity. That statement is not in the reference ratio and is not correct, because an easement of necessity does not depend on a period of use. It is an example of the model stating wrong law confidently.

6. Plain-language rewriting

The version 2 ratio was rewritten for a general reader using a second prompt: explain the principle in 3 to 4 simple sentences a Class 8 student could understand, keep section numbers and figures exactly as written, and use only what is in the text. All 20 rewrites were saved to the outputs folder.

> **Extracted ratio (version 2):** An easement by prescription under Section 15 of the Indian Easements Act, 1882, requires continuous and uninterrupted use as of right for twenty years ending within two years prior to the suit. In cases of necessity, however, a shorter period may suffice, and the court can grant an easement if the user is necessary for the enjoyment of the dominant tenement. Here, the plaintiff proved an easement of necessity due to the land's inherent landlocked condition, necessitating an alternative route for ingress and egress.
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> **Plain-language version:** According to the law, if someone uses another person's land without permission for 20 years, they might get a special right to use it. But if they need to use it because their land has no way to get to other places, they can use it even if it's not 20 years. The person who needed to use the land had no other way to reach their property easily, so the court allowed them to use it.

Observations:

- The language is simple and easy to follow, which is the goal of the project.
- The rewrite repeated the error from the extracted ratio ("even if it's not 20 years"), so simplifying does not remove mistakes.
- The instruction to keep section numbers was not followed: "Section 15 of the Indian Easements Act, 1882" disappeared. A later step should attach statute references back to the explanation.
- "Without permission" is a loose way of saying "as of right", which shows how plain wording can blur a legal term.

7. Findings

1. A prompt-based 3B model can identify the central legal distinction in a case, but it does not reproduce all the principles, statute sections and named tests of the reference ratio.
2. Two worked examples made the output more like a real ratio (direct opening, statute citation, formal tone) and gave a small gain in BERTScore and ROUGE-L.
3. The model can state incorrect law with confidence, and the plain-language step carries such errors forward.
4. These points support the later phases: grounding legal terms and sections from a verified glossary, and a hallucination guard that checks each claim against the source text.

8. Limitations

- Only 20 test cases were scored, so small score differences may be noise.
- The data is synthetic and uniform, so results may look better than on real judgments.

- The reference ratio for the case studied also contains principles (equity, interest) that do not obviously relate to an easement dispute. This should be verified against the full case, because it would cap the scores that any model can reach.
- Only one case was compared in detail, and the plain-language rewrites were judged by reading, not scored.

9. Next steps

- Read three or four more predicted ratios next to the reference ratios and record the common gaps.
- Optional improvement: choose the two examples for each test case by similarity (matching legal terms or issues) instead of using two fixed examples.
- Phase 3: build the legal-term glossary (31 unique terms in the dataset) with retrieval, and add translation into Hindi, Kannada and one more Indian language.
- Phase 4: add the hallucination guard and the final evaluation and demo.

10. Deliverables

- Notebook: 01_phase1_and_2_cleaning_and_baseline.ipynb (Phase 2 cells: ratio extraction, scoring, plain-language rewriting)
- outputs/ratio_predictions.jsonl (version 1 predictions)
- outputs/ratio_scores.csv (version 1 scores)
- outputs/ratio_predictions_v2.jsonl (version 2 predictions)
- outputs/ratio_simple.jsonl (plain-language rewrites)